TexCycle AI: An AI-Powered Textile Waste Marketplace and Circular Economy Intelligence Platform for Sri Lanka

INDIVIDUAL THESIS

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Individual component: Sustainability Intelligence

Group ID: R26-IT-081

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) in Information Technology

Specializing in Information

Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Textile waste is an environmental burden and a circular value creation opportunity. The Member 02 Sustainability Intelligence component of the larger TexCycle AI marketplace offers a data-driven decision layer to quantify the environmental and circular value of textile materials and to drive the right recovery actions. The component accepts textile and condition attributes such as fabric type, weight, health score, repairability, condition, district, province, and industry type. Six XGBoost regression estimators were used to predict circularity, recyclability, percentage of carbon-reduction, percentage of water-reduction, CO 2 savings and water savings. The six regression outputs, original inputs and derived sustainability indicators are consumed in a separate stage of grade-classification. The implementation also provides scenario comparison, risk assessment, explainable recommendations, impact-factor versioning, prediction history and integration outputs for downstream marketplace services.

The development followed a workflow of data preparation, pre-processing, model training, evaluation, model artifact, API integration, and application testing. The reported development evaluation produced R2 values of 0.9657 for circularity, 0.9502 for recyclability, 0.9589 for carbon reduction, 0.9389 for water reduction, 0.9887 for CO2 saving and 0.9854 for water saving. These results indicate good fit on available development evaluation data but should not be interpreted as certified environmental accuracy since the current sample dataset is synthetic and rule generated. The last contribution is therefore a reproducible sustainability-intelligence workflow and integrated software component, rather than a claim of real-world environmental measurement accuracy. Validation requires real Sri Lankan textile-waste observations, calibrated life-cycle inventory factors and independent evaluation.

Keywords: textile waste, sustainability intelligence, circular economy, XGBoost, environmental impact

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to the academic staff and research supervisors at Sri Lanka Institute of Information Technology (SLIIT) for their guidance, valuable feedback, and continuous support during the development of the Sustainability Intelligence component of the research project.

I am also grateful to my research team members for their collaboration in developing the integrated textile-waste marketplace. Their contributions to the Fabric Intelligence, Marketplace Predictive Analytics, and Commercial Intelligence components enabled the integration of sustainability predictions into the wider platform.

Finally, I acknowledge the research publications, open-source libraries, technical resources, and development tools that supported the implementation, experimentation, and documentation of this work.

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
API	Application Programming Interface
CO_2	Carbon dioxide
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
FAPI	FastAPI framework
LCA	Life Cycle Assessment
MAE	Mean Absolute Error
ML	Machine Learning
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

The research project addresses the opportunity of treating textile waste as a source of recoverable circular value, instead of as an undifferentiated disposal stream. The larger platform, called TexCycle AI in the research materials, combines textile intelligence, sustainability intelligence, commercial intelligence, marketplace prediction and traceability into one digital workflow. The context is the Sri Lankan textile and apparel sector where textile materials may have value even when they are no longer suitable for their original purpose. A central motivation for the integrated platform is fragmentation between material identification, environmental assessment, commercial valuation, buyer discovery and lifecycle traceability [1, 2, 3] according to the research draft.

The Sustainability Intelligence component is one of the main decision layers in that platform. Its purpose is not simply to give a "green score" to someone. Instead, it converts structured textile characteristics to a set of environmental and circularity indicators, assesses alternative recovery scenarios, detects risks and provides an explainable recommendation consumable by the marketplace. This position is consistent with circular-economy thinking, where reuse, repair, remanufacture and recycling are considered in relation to product life extension and resource efficiency [2, 4, 5].

1.2 Individual component context

02 Sustainability Intelligence Member. The component receives information on a textile item and produces environmental decision support outputs. The documented implementation takes eight primary inputs which are fabric type, weight in kilograms, health score, repairability, condition, district, province and industry type. These inputs are used to feed a sustainability prediction pipeline with six regression targets and a

downstream sustainability-grade classifier. The software implementation also has deterministic services around the ML layer. The services deals with environmental impact factors, risk assessment, explanations, scenario comparison, report summaries and integration payloads. This architecture is important as the research does not consider a model prediction to be a complete truth of an environment. The model is one part of an auditable workflow for decision support.

1.3 Problem statement

When making decisions about textile waste, material information, condition information and environmental information are not connected. A textile item may be technically recyclable but not economically viable for a given pathway; a material with high potential for environmental savings may need to be repaired or cleaned before reuse; and a long transport distance may undermine the practical attractiveness of a recovery option. Hence, the larger problem posed in the research draft is how to integrate artificial intelligence, sustainability analytics and predictive marketplace technologies into a unified Sri Lankan textile-waste marketplace [3, 6].

The challenge for Member 02 is to translate available textile and condition characteristics into consistent sustainability indicators and actionable recovery recommendations while preserving explainability and accounting for uncertainty in the underlying environmental assumptions.

1.4 Research gap

The available literature and the draft for the project indicate a few research positions that are only indirectly linked. Research on textiles focuses on classification based on materials; recycling studies investigate sorting and environmental aspects; work on digital passports concentrates on the issue of traceability; marketplace research studies buyers and sellers; and price or recommendation systems look at commercial

decisions. The gap that is significant for Sustainability Intelligence is that there is not enough environmental decision support integrated with commercial activities and traceability in a textile marketplace [4, 7, 6, 8].

A further gap is geographical. The existing studies on textile circularity rely extensively on datasets and life-cycle information from countries other than Sri Lanka. This means that our current study treats Sri Lankan textile waste data as the future validation requirement and not automatically assuming the transferability of external information.

1.5 Aim and objectives

The primary purpose of the study is to design, devise and carry out a study on a marketplace that uses artificial intelligence to conduct an analysis of textile waste in Sri Lanka and determine the textile properties; the marketplace estimates environmental costs and commercial potential, as well as makes recommendations by indicating buyers and enforcing tracking of textile lifecycle. The mission of Sustainability Intelligence itself is to establish a decision-making process for sustainability that will be developed in such a way that it can be replicated and produce useful results.

- Determine input variables for sustainability assessment and define their representation.
- Create a multi-output regression pipeline model for circularity, recyclability, carbon reduction, water reduction, CO_2 saving, and water saving.
- Introduce a classification stage using the output from the model and indicators derived.
- Combine the sustainability pipeline with Fast API service and a React-based seller workflow.
- Compare scenarios, assess risk, and provide interpretability of the ML output.
- Assess the pipeline’s work using regression and classification metrics.

- Indicate limitations of the work done and plan next steps for validation by using Sri Lankan textile-waste real data.

1.6 Research questions or hypotheses

RQ1: Is it possible to convert structured apparel and condition attributes into relevant sustainability indicators via a replicable machine-learning process?

RQ2: Is it possible to obtain different environmental targets within a comparable sustainability-intelligence framework by keeping preprocessing identical during training and inference?

RQ3: Is it feasible for a sustainability-quality classification system to utilize the predicted environmental indicators to create useful recovery solutions?

RQ4: Is it feasible to incorporate sustainability results into a marketplace workflow to enable environmental intelligence to inform downstream pricing, matching, and listing decisions?

H1: A structured multi-outcome sustainability pipeline can reach acceptable predictive performance by testing it against the available development data using MAE, RMSE, and R^2.

H2: The integration of determinism around the machine learning component makes decision-making process richer than just demonstrating one raw score of the model.

1.7 Scope limitations and assumptions

The functional description includes the Sustainability Intelligence module and its additional features related to the market. It considers topics such as environmental forecasting, sustainability assessment, comparison of scenarios, determination of risks, historical information tracing, and other integration impacts. There is no intention to substitute a real-life LCA process, environmental product declaration or an independent accounting of carbon footprint. One should note that dataset generated for the project is synthetic and modeled for validation purposes; therefore, the numerical analysis results cannot be considered as the evidence that real environmental savings can be identified correctly.

The implementation assumes that the information concerning textile properties provided by upstream modules is sufficient for the sustainability model. If some data are missing or dubious, the distances will indicate the poor quality of data or the risk level of encountered results.

1.8 Individual contribution

The timeline of contributions made indicates that considerable efforts have been made in order to carry out requirements analysis and sustainability literature review before engaging in data collection, training of models, comparing the model, tuning it, performing API integration, integrating the system into the frontend, testing and final evaluation. Interests mainly focused on finding the environmental indicators and studying methods for reducing carbon dioxide level and minimizing the usage of water. The next step included activities related to organizing data and cleaning it in the course of training and comparing the models. Further, the next steps had to do with the integration of models and testing seller workflow.

1.9 Thesis structure

Chapter 1 gives an overview of the research context, problem, gap, objectives and contribution. Chapter 2 gives a review of the foundations of the area of study, relevant studies and techniques available in the technical literature and creates the background for the gap that is tackled in the component. Chapter 3 describes the research methodology including the design, data, experiments, assessment, analysis of the experimental data, ethics and reproducibility of the research. Chapter 4 discusses the specifications for the product, its architecture, design choices, detailed sustainability pipeline and requirements. Chapter 5 describes the implementation, integration, testing and deployment process. Chapter 6 concerns the findings of the model and the system. Chapter 7 interprets these findings and correlates them with the literature. Chapter 8 discusses the commercialization and sustainability. Finally, Chapter 9 summarizes the research and suggests some directions for further work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review examined five main domains: textile waste and circular economy; environmental assessment and life cycle thinking; artificial intelligence for textiles analysis; digital traceability and circular marketplaces; and machine learning methods applicable to sustainability predictions. The research draft provided an outline of the literature structure for the project while established environmental and methodological data sources were referenced for providing the necessary explanations on each of the aforementioned agendas. [2, 4, 9, 10] .

The review attempted to solve various problems raised by each of the identified agendas, such as the type of data needed in each case and the elements of those directions that do not link to the decision-making process in the business environment. It should be noted that the distinction between actual environmental measurements and projections is also crucial because a model may be able to extract patterns from a given dataset, but it would not work as a valid environmental measurement tool.

2.2 Domain and theoretical foundations

2.2.1 Circular Economy and Textile Waste

Strategies associated with a Circular Economy aim to promote repairing, reusing and recycling of products and thus reduce the use of primary materials. In particular, the textile industry applies these principles by offering a number of options in terms of action, namely reuse, repair, upcycling and recycling, with disposal being the least favorable path. Ellen MacArthur Foundation’s textile project promotes the idea of redesigning the whole system in such a way that clothes and textile materials can be used in circulation for extended periods of time. Similarly, reviews of textile waste management indicate the fact that the aforementioned issues are interrelated [4]. Thus, according to the circular economy framework, the Sustainability Intelligence module

establishes certain grounds for making a recommendation based on the recovery pathways rather than making a decision of the immediate waste classification. In addition, it needs to be emphasized that the decision support system should take into account the material type, condition, expected environmental outcome and its overall feasibility.

2.2.2 Life Cycle Assessment and Environmental Indicators

Life Cycle Assessment has been developed into a methodological approach that allows to assess environmental characteristics of a product system. ISO 14040 defines the basic principles and framework for LCA, and ISO 14044 defines requirements and guidelines for the study of LCA in practice [10, 9] In contrast to that, the current scenario does not execute a full ISO-compliant LCA for each textile item. Instead, it uses structured parameters and assumptions that have been informed by LCA and results in environmental indicators that are connected with stages of product development.

It is important to maintain this distinction in the system documentation. The interpreted CO_2 and water figures should be understood as modelled indicators that are created based on given assumptions. Furthermore, the next stage of the research should enable to link the platform with the reliable inventory datasets, certain impacts concerning the regions, estimation ranges, and the definitions of functional unit.

2.2.3 Artificial Intelligence in Textile Sorting

Research on textiles using artificial intelligence has included image classification, fibre identification, defect detection, and automated segregation. The preliminary research articulation mentions EfficientNet-B3 for the achievement of 16-category fabric classification and MobileNetV2 for binary flaw identification in broader Textile Intelligence. These earlier outcomes are relevant for Sustainability Intelligence since

the type and condition of materials influence environmental routing options. Research on fabric classification also indicates that blended fibers, limitations of sensing, and availability of data continue to be problems [11, 12, 13, 14, 15, 16] .

2.2.4 Machine Learning for Structured Prediction

Sustainability pipeline uses XGBoost for tabular prediction due its performance over non-linear relationships and due nature of mixed characters of variables . The current framework incorporates separate regression models due to the different units and meaning for targets, followed by separate classifier that evaluates sustainability grade.

The choice of the model is based on the applicability of XGBoost and practical requirements of the project rather than the assumption that this is the best model ever identified.

2.3 Related research

Repeatedly, research on textile circularity identifies data quality as a significant impediment to credible environmental assessment. According to literature concerning textile circularity and life cycle assessment (LCA) applications, various methodologies, unavailability of initial data, and ambiguity when comparing reuse vs. recycling options are present [4, 6, 8, 17] . Being involved in this research clarifies the rationale behind unifying snapshots of environmental factors when making predictions.

Research on AI-base textile sorting confirms that machine learning technology enables the capability of transforming visual or spectral data into relevant material information [11, 11, 12, 11, 18, 14, 13] . However, knowledge of material identification is insufficient to define the optimal marketplace behavior from the economic or environmental perspective. Thus, Sustainability Intelligence is positioned between the textile identification and optimization in marketplace activities.

2.4 Existing systems products and techniques

These existing techniques may be divided into material recognition techniques, environmental assessment techniques, traceability techniques, and marketplace techniques. Material recognition techniques rely on high levels of classification, whereas LCA techniques consider the environmental impact within system boundaries. Digital product passport techniques are primarily concerned with preservation of product data throughout its life cycle, and marketplace techniques revolve around listing, discovery, pricing, and trading. The novel technique is different from these existing techniques because it combines all these decision-making needs. It takes input in the form of textile data, predicts sustainability parameters, analyzes different recovery scenarios, captures the environmental context underlying the predictions, and outputs something clearly useful for marketplace components.

Research direction	**Primary strength**	**Remaining limitation relevant to this work**
Textile AI sorting	Automated material/fiber recognition	Does not by itself determine marketplace recovery value.
NIR/CNN recognition	Detailed fiber identification	Requires specialized sensing and validated data.
Textile sustainability/LCA	Environmental assessment framework	Often separated from marketplace decision-making.
Digital product passports	Lifecycle traceability	Not necessarily an AI-enabled circular marketplace.
Circular marketplaces	Buyer/seller connection	Limited integrated sustainability prediction.
Price prediction	Commercial valuation	Usually not coupled directly to environmental intelligence.
Recommendation systems	Matching users and items	Textile-specific circularity context is often limited.

Table 1. Comparison of related research directions and the identified gap

2.5 Gap synthesis and conceptual framework

There is a need for a sustainability decision layer, which needs to be associated with, but distinct from, the larger marketplace based on the synthesis of literature. The conceptual model used in this thesis is as follows:

Textile properties → standardized properties → six sustainability regressors → indicators generated → sustainability rating → risk/explanation → scenario comparison → inclusion in the marketplace.

Moreover, the conceptual model allows for a loop back for subsequent validation, whereby real-life transaction and recycling results can be used to further develop environmental parameters and the prediction models. This process develops into a progression from an initial machine learning model to a circular marketplace.

Layer	**Inputs**	**Outputs**	**Purpose**
Material/condition layer	Fabric, weight, health, repairability, condition	Standardized assessment record	Define the item.
Prediction layer	8 raw features	6 environmental/circular outputs	Estimate sustainability indicators.
Derived-indicator layer	Predictions + known inputs	Energy, landfill, SDG, economic indicators	Create decision context.
Grade layer	18-feature representation	Sustainability grade	Summarize suitability.
Decision-support layer	Grade + risk + factors	Recommendation, explanation, scenario ranking	Support action selection.
Marketplace layer	Sustainability output	Pricing/matching inputs	Connect environmental value to commerce.

Table 2 Conceptual Sustainability Intelligence framework

3 RESEARCH METHODOLOGY

3.1 Research design

The research uses an applied design-and-evaluation research methodology. The research problem is formulated in the form of a software artifact, created via reproducible machine learning and API. The assessment of models and the behavior of the system happens by way of metrics and software tests. Iteration is the key characteristic of the methodology: requirements definition, data preparation, model training and comparison, pipeline integration, and testing of the integrated system take place.

The method defines three levels of evidence: model-level evidence, software-level evidence, and interpretation in research level. The former includes metrics of predictive power of models in the available evaluation set; the latter is about the conformance of the implementation with its intended behavior. Neither level by itself implies environmental validity.

3.2 Data participants or study materials

The main data used in the ML pipeline under consideration is the structured dataset on textile sustainability that serves as data both for development and testing purposes. It is artificial and created using rule-based techniques, and its purpose is to check data flow, feature engineering, training, model persisting, and inference. As a result, the data can be considered development data but not representative data on Sri Lankan textile waste.

Moreover, the project research draft introduces a prototype marketplace dataset that includes 36 features and 100,000 artificial entries to be used in marketplace prediction. It is part of the bigger platform and is not the basis for Member 02 environmental model. Sustainability Intelligence evaluation is limited only to six regression outputs and grade classification process.

Feature group	Variables	Role
Raw categorical	fabric_type, repairability, condition, district, province, industry_type	Describe material and context.
Raw numeric	weight_kg, health_score	Describe quantity and condition.
Regression targets	circularity_score, recyclability_score, carbon_reduction_percent, water_reduction_percent, co2_saved_kg, water_saved_liters	Environmental/circular predictions.
Derived indicators	energy_saved_kwh, landfill_diverted_kg, sdg_impact_score, economic_impact_score	Decision-support context.
Classification representation	8 raw + 6 predicted + 4 derived = 18 features	Input to grade classifier.

Table 3 Sustainability Intelligence feature and target structure

3.3 Experimental or study procedure

The documented machine learning process follows the order below: data acquisition, data cleaning, exploratory validation, feature engineering, split into training and test sets, preprocessing, baseline modeling, model training, hyperparameter tuning, model evaluation, choosing the best-performing model, persisting the model, integrating it into an API, and finally testing the application. The order was selected to guarantee the traceability of the transformation from research to production.

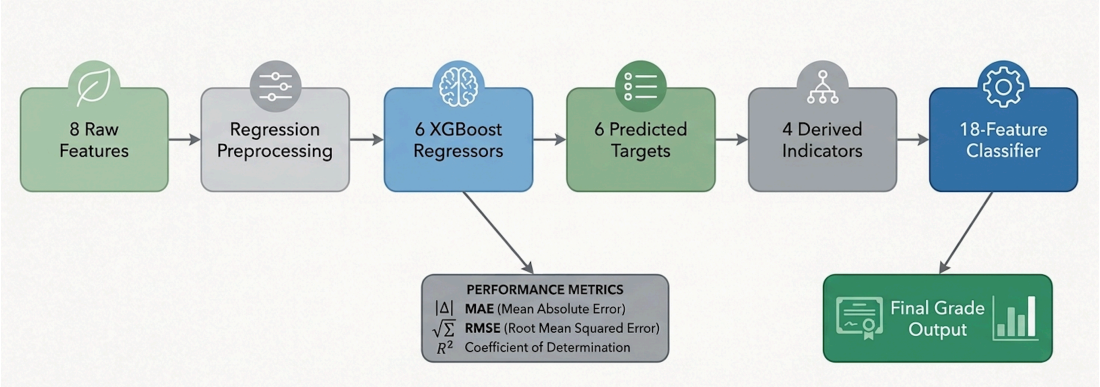


Figure 1.Multistage sustainability assessment inference pipeline

The inference approach follows the steps of the training process. There are eight raw features that need to be transformed using a specialized regression preprocessing pipeline. There are six XGBoost regressor models whose outputs serve as the environment data, and which are combined with the raw input features and four other calculated indicators into a representation of an eighteen-feature classifier. There is a separate preprocessing object used for transforming the classifier representation before the grade is predicted.

Steps:

1. Verify the input schema and normalize categorical values.
2. Provide the eight raw sustainability features.
3. Transform the features using the regression preprocessing pipeline.
4. Predict using six XGBoost regressors.
5. Calculate indicators based on predictions and input features.
6. Create the classifier representation.
7. Deploy the custom preprocessing workflow for the classification model.
8. Produce the sustainability grade forecast and interpret the respective grade code.
9. Carry out clipping, validation, and response formatting.
10. Distribute the output through the risk assessment, explanation, and integration service flows.
11. Save the forecast along with the snapshot of the environmental factors.

3.4 Measures and evaluation metrics

Regression models will be judged by MAE, RMSE, and R^2. MAE is used to calculate the mean absolute prediction error in the same unit as that of the target variable. RMSE places more emphasis on larger errors. R^2 refers to the ratio of variance explained against the evaluation baseline. The metrics are to be understood as a collective.

Accuracy, macro precision, macro recall, macro F1-score, and a confusion matrix are the metrics for evaluating the grade classifier. Other system metrics are smoke tests for the API, frontend tests, model loading checks, and builds. Considering the synthetic nature of the dataset, the metrics serve as development/pipeline validation metrics.

Metric	Interpretation	Why used
MAE	Average absolute error	Easy-to-interpret error in target units.
RMSE	Square-rooted mean squared error	Highlights larger prediction errors.
R^2	Variance explained by model	Measures relative predictive fit.
Accuracy	Correct classification proportion	Basic grade/action correctness.
Macro precision	Average class precision	Reduces dominance of large classes.
Macro recall	Average class recall	Checks class coverage.
Macro F1	Balance of precision and recall	Useful for multi-class comparison.
Confusion matrix	Class-by-class error pattern	Shows which grades/actions are confused.

Table 4 Evaluation metrics used for the Sustainability Intelligence component

3.5 Data analysis

Data analysis involved evaluation of missing values, categorical outliers, distribution of targets, and dependency of input variables on predicted variables. Error analysis was conducted during the modeling process in order to detect weak points and to guide preprocessing and hyperparameter selection. The final analysis will focus on how well the pipeline satisfies the technical goals, and not whether the high R^2 scores are true representations of the environment.

In case of an upcoming real-world data study, the analysis should be extended to incorporate stratified analysis based on fabric type, fabric conditions, district and industry type, calibration of confidence intervals, analysis of residuals, outliers, validation based on time or location, and assumption sensitivity analysis based on environment.

3.6 Ethics privacy security and responsible AI

The current data set being used for the development phase is artificial, thus minimizing any privacy concerns that might arise in the testing phase. In case of an actual data set in future, it may contain sensitive data regarding the amount, locations, manufacturing process, or supplier of textiles. These data need to be minimized, access controlled, and anonymized or pseudonymized whenever possible. Responsible AI is especially important here due to the possibility of environmental scores affecting the decision-making process in businesses. The system needs to state the source of the model, confidence level or quality of data, and contextual data used to make a prediction.

3.7 Reproducibility and research artefacts

Models are stored in JSON format using native XGBoost models, while joblib is used only for pre-processing and label-encoding artifacts. Such a decision is made to avoid possible problems connected with serialization of the inner structures of XGBoost. The part of inference uses native JSON boosters and different fitted preprocessors for regression and classification.

Reproducibility is achieved by recording the versions of packages, rules for the creation of the dataset, feature names, target names, model version, and factor version. Future releases must include immutable experiment identifiers and automated model cards to allow traceability of each deployed model to their training and testing datasets.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Sustainability Intelligence module was designed in line with functional as well as non-functional requirements. Functional requirements include the acceptance of assessment data, making sustainability predictions, grading of materials, scenario comparison, result storage, displaying of historical data, impact factors management, and exposing of results to downstream marketplace modules. Non-functional requirements include explainability, consistency between training and inference, responsive user interface behavior, error management, and auditability.

ID	Requirement	Priority	Verification
FR-01	Accept validated textile and condition inputs.	High	Assessment API test
FR-02	Predict six sustainability indicators.	High	Model inference test
FR-03	Generate sustainability grade.	High	Classifier test
FR-04	Compare reuse, repair, upcycle, recycle and discard.	High	Scenario test
FR-05	Provide explanation and risk indicators.	High	Service test
FR-06	Persist predictions and factor snapshots.	High	Database test
FR-07	Provide history and dashboard analytics.	Medium	Frontend/API test
FR-08	Expose downstream integration payload.	High	Integration test
FR-09	Manage datasets and model registry.	Medium	ML management test
NFR-01	Maintain training/inference preprocessing consistency.	High	Artifact verification
NFR-02	Provide graceful fallback when trained artifacts unavailable.	Medium	Failure-path test
NFR-03	Prevent unintended horizontal UI overflow.	Medium	Responsive UI test

Table 5 .Functional and non-functional requirements

4.2 Component architecture within the complete solution

The wider platform contains a React/Vite frontend, Node.js/Express application services, Python/FastAPI ML services, database-backed storage and marketplace intelligence. Member 02 is implemented as a sustainability decision layer within this architecture. The frontend collects assessment information and displays results. The FastAPI service validates the request and orchestrates model inference. MongoDB Atlas stores inputs, predictions, scenarios, impact factors, datasets and model metadata.

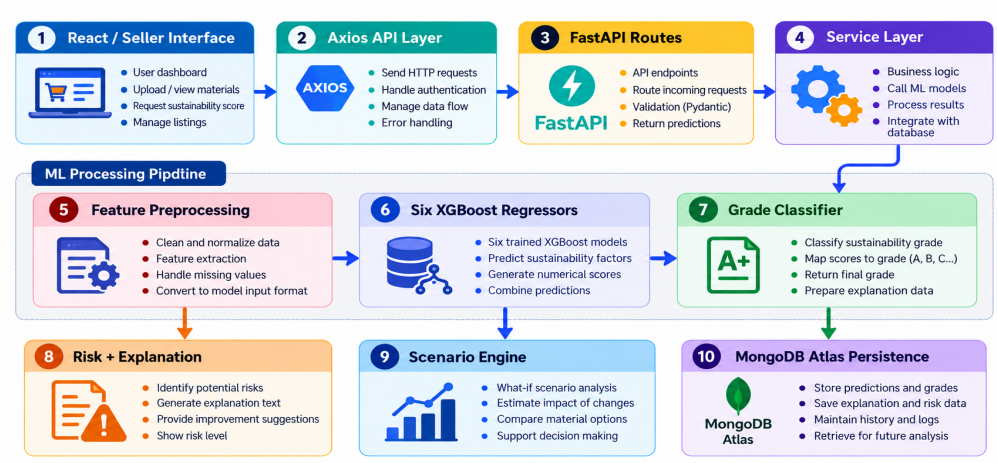


Figure 2:Sustainability Intelligence Architecture

This border is deliberately constructed to separate predictions about the environment from decisions made by market actors. The sustainability service is not responsible for setting the price or involving consumers directly but rather provides the sustainability information structure that can be used by the subsequent components.

Layer	Technology / component	Member 02 responsibility
Presentation	React, TypeScript/JavaScript, Tailwind CSS	Assessment, results, dashboard, history and ML management UI.
API	FastAPI	Validation, prediction and integration routes.
Service	Python service modules	Risk, explanation, scenario, analytics and report orchestration.
ML	XGBoost + scikit-learn	Six regressors and sustainability-grade classifier.
Persistence	MongoDB Atlas / Motor	Inputs, predictions, factors, datasets and model registry.
Integration	REST/JSON payloads	Expose sustainability results to marketplace components.

Table 6Technology architecture for Sustainability Intelligence

4.3 Design alternatives and decisions

Rules-based modeling appeared appealing due to its transparent assumptions regarding the environment. However, this problem required an investigation-based machine learning element capable of learning structured connections from the training set. While using a single multi-target estimator was considered, it was decided to utilize discrete regressors as each of the six targets has different units and properties of errors.

In terms of the preservation of artifacts, native JSON was used for the preservation of the XGBoost model instead of serialization through joblib. At the preprocessing step, two fitted ColumnTransformers were preserved instead of having a mutable object shared between the regression and classification parts of the pipeline. For the purposes of the stable release, the API allows the use of placeholder, trained, and auto models.

4.4 Detailed design

4.4.1 Input and Feature Design

These eight raw inputs are specifically chosen to fit in with the workflow of the seller. The fabric type represents a material category. The weight is the quantity. The health score and condition represent the state of the item. Repairability represents whether the item can have its lifespan extended. District, province, and industry type provide additional context, which can affect logistics or patterns in certain industries.

The four new inputs are then added to the six predicted values to form four more features: energy saved, landfill saved, SDG impact, and economic impact. As such, the classifier uses eighteen features in total. This implementation maintains the names of the raw inputs and outputs explicitly within the model code itself.

4.4.2 Environmental Factor Versioning

Environmental factors are stored as versioned records. Each prediction saves both the factor version and a factor snapshot. This is an important audit mechanism: if an administrator changes a factor later, the historical prediction still records the factor values that were used at prediction time. The design is aligned with the broader LCA literature's emphasis on transparent assumptions and data quality [4, 17, 10, 9] .

4.4.3 Scenario Decision Design

The scenario engine analyzes the potential for reuse, repair, upcycling, recycling, and disposal. In contrast to the evaluation of scenarios by their environmental performance alone, environmental performance, feasibility, difficulty, and risk are considered together. This allows avoiding the recommendation of a scenario with a high environmental score theoretically, while being practically infeasible because of the state of the textile.

The result is a set of scenarios with environmental performance, feasibility, difficulty, score, risk, badge, and explanation. The best feasible scenario is recognized as the recommended scenario. The whole process is designed to be very transparent and easily calibrated later based on experience.

4.4.4 Explainability and Risk Design

The explanation service provides the main reason, supporting reasons, negative aspects, and suggestions for improvements. The risk service is able to recognize states such as excessive contamination, long transport distance, poor quality, considerable damage, unusual weight, and unknown fabric material. The data quality state is provided along with the forecast in order to distinguish whether the result is reliable or not.

4.5 Standards and Professional Practice

The interpretation of environmental issues agrees in principle with ISO 14040 and ISO 14044 standards but acknowledges that the prototype cannot be treated as a complete LCA study. The tree boosting method is as documented by Chen and Guestrin. Software best practices emphasize modular pathways, schema verification, error control, version evolution, and separating prediction activities from marketplace decision-making.

The thesis also follows SLIIT's dissertation formatting guidelines, specifically Times New Roman font, required heading sequence, 1.5 line spacing of the main text, required margins, chapters-wise figure and table numbering, Roman format pagination for the preface, and Arabic format pagination for the body of the dissertation.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The software consists of a frontend based on React and a backend that uses Python/FastAPI together with MongoDB Atlas for data storage. The ML layer takes advantage of several Python frameworks like pandas, scikit-learn, joblib and XGBoost. It also integrates Axios in order to handle communication through APIs and employs Tailwind CSS for UI design. The documentation describes the implementation in details, covering model management, dataset upload and report summary services.

The software is designed in a way so that the user interface does not execute the main sustainability prediction task. Instead, the UI state is transferred into the request format used in the backend, which takes care of selecting the model and applying it for making predictions. Finally, results are sent back to the UI.

Technology	**Use in Member 02**
React	Assessment, dashboard, scenario and history interfaces.
Tailwind CSS	Responsive UI layout and visual styling.
Axios	Frontend/backend API communication.
FastAPI	Prediction, scenario, history, factor, dataset and model routes.
Pydantic	Request validation and API schema enforcement.
MongoDB Atlas	Persistence of predictions, factors, datasets and model metadata.
Motor	Asynchronous MongoDB access.
pandas	Tabular data preparation and feature handling.
scikit-learn	Preprocessing and evaluation utilities.
XGBoost	Six regression models and grade classifier.
joblib	Persistence of preprocessing and encoding artifacts.

Table 7.Development technology stack

5.2 Component implementation

5.2.1 Sustainability Model Pipeline

According to the implementation, the term RAW_FEATURES refers to fabric type, weight, health score, repairability, condition, district, province, and industry type.

The regression preprocessor, classifier preprocessor, and grade encoder are loaded via jilbab, at the time of code execution. Each of the XGBoost boosters is loaded using loading method offered by the technology itself. The execution of the six regressors comes first in time, and classifiers are executed later, following the established design of training, where the classifier refers to the regressors' predictions instead of ground-truth values.

5.2.2 API Implementation

The key endpoint for sustainability functionality is POST /api/sustainability/predict as outlined in the Eco Thread documentation. The process includes the validation of the request made, impact factors selection and the prediction model activation, output enrichment through risk and explanation data, result persistence, and the final structured response generation. It should also be mentioned that apart from the endpoint named, in the broader seller flow grading is facilitated by the use of the ML layer, and the resulting sustainability indexes are transferred to marketplace forecasting.

The system was created so that if the trained artifacts do not load properly, the whole application does not become unavailable. In automatic mode, the loader is supposed to revert to the placeholder model. The said feature constitutes a specific feature aimed at providing reliability and should not be seen as a replacement of the production model.

5.2.3 Frontend Implementation

The frontend has numerous components such as Sustainability Assessment, Scenario Simulation, Dashboard, Prediction History, Impact Factors and ML Models views. The reusable components also have page titles, KPI cards, tables, forms, charts, recommendation labels and report summaries. The concerned implementation deals with the responsive-layout aspect of Scenario Simulation by managing the five-column grid to work only on wide screens and avoids the problem of horizontal overflow.

On the seller side, the upload process can call the sustainability evaluation service after performing the textile assessment at the beginning of the process. The implementation allows passing the specifics about the fabric type, weight, health level, repairability, condition, district, province, and industry type during the evaluation process. Eventually, the sustainability output may be fed to the marketplace prediction engine.

5.3 Integration with the other components

Integration is based on a clean sustainability output contract. Downstream pricing can consume circularity, recyclability, carbon reduction, water reduction, CO2 saving and water saving. Marketplace matching can consume recommended action, distance related context and marketplace priority information. The seller workflow therefore follows a chain in which image based textile analysis provides material and condition information, Sustainability Intelligence provides environmental intelligence and marketplace intelligence uses the combined information for commercial decisions.

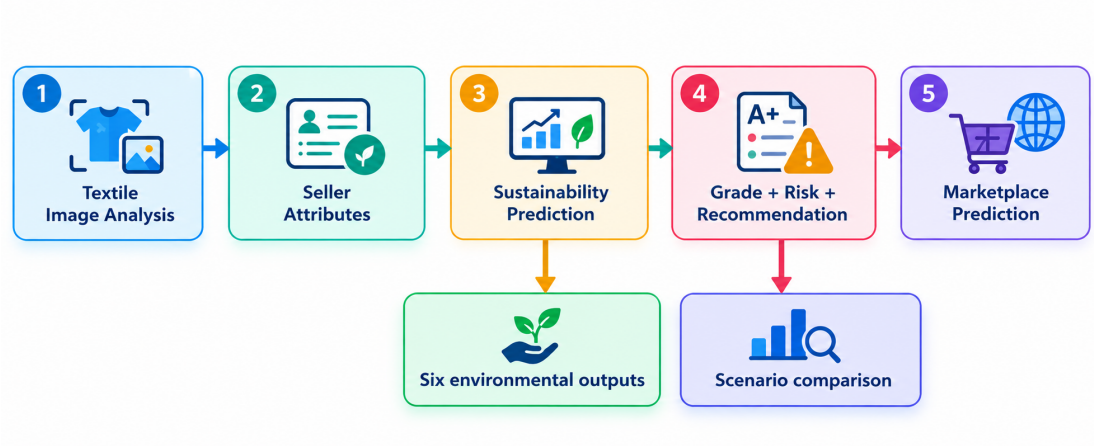


Figure 3:Seller-to-Marketplace Sustainability Integration

This integration preserves component ownership. Member 02 does not implement buyer recommendation or final price prediction; it supplies sustainability features that those modules can use.

5.4 Verification and software testing

It is necessary to perform testing on the model, service, API, frontend, and build levels. Tests on the model level check if it is possible to load the expected set of feature names and artifacts. Service tests check if risk, explanation, and scenario calculations are correct. API smoke tests ensure that all routes accept the valid input and provide the expected response structure. Frontend tests check API mapping and sustainability calculation helpers. Finally, a build verification test ensures that the React app builds correctly.

The manual verification process includes launching the backend and frontend, checking the health route, submitting a sustainability assessment, checking out the results and risk indicators, performing a scenario simulation, inspecting dashboard totals, filtering the prediction history view, and checking dataset/model-management features. Such an approach covers more ground than a single-snapshot automated test, as it examines the application's logic and frontend components.

Test area	Example test	Expected outcome
Model loading	Load all six XGBoost regressors and classifier	All required artifacts load without serialization errors.
Feature consistency	Send valid eight-feature assessment	Prediction returns six environmental outputs.
Classifier flow	Run grade prediction	Classifier receives regressor predictions and returns a valid grade.
Scenario engine	Compare five recovery actions	Best feasible action is identified with score and explanation.
Risk service	Submit high contamination/long distance case	Appropriate risk flags and data-quality status are returned.
History	Save prediction then query history	Saved record is retrievable with correct metadata.
Integration	Request sustainability output payload	Downstream-ready JSON is returned.
Frontend build	Run production build	Build completes without compilation errors.

Table 8 .Representative verification tests for the component

5.5 Deployment and operation

The architecture was designed in a way that decouples the frontend, backend, and ML components, so the sustainability service can be deployed independently when needed. The environment configuration defines the database connection, the frontend origin, the model mode, and the model version. The model artifacts should be delivered with all the preprocessing and label-encoding steps, which are expected by the inference code.

The operational monitoring should store the model version, factor version, the prediction timestamp, input quality indicators, and model source for auditability and later model retraining. The user authentication and authorization are features that should be prioritized in the future, as the current design is aimed at research demonstration and production-ready deployment.

6 RESULTS AND VALIDATION

6.1 Research results

The current documented development evaluation reports strong R^2 values across all six sustainability regression targets. Circularity prediction achieved $R^2 = 0.9657$; recyclability prediction achieved $R^2 = 0.9502$; carbon reduction achieved $R^2 = 0.9589$; water reduction achieved $R^2 = 0.9389$; CO_2 saving achieved $R^2 = 0.9887$; and water saving achieved $R^2 = 0.9854$. The corresponding MAE and RMSE values are reported in Table 6.1.

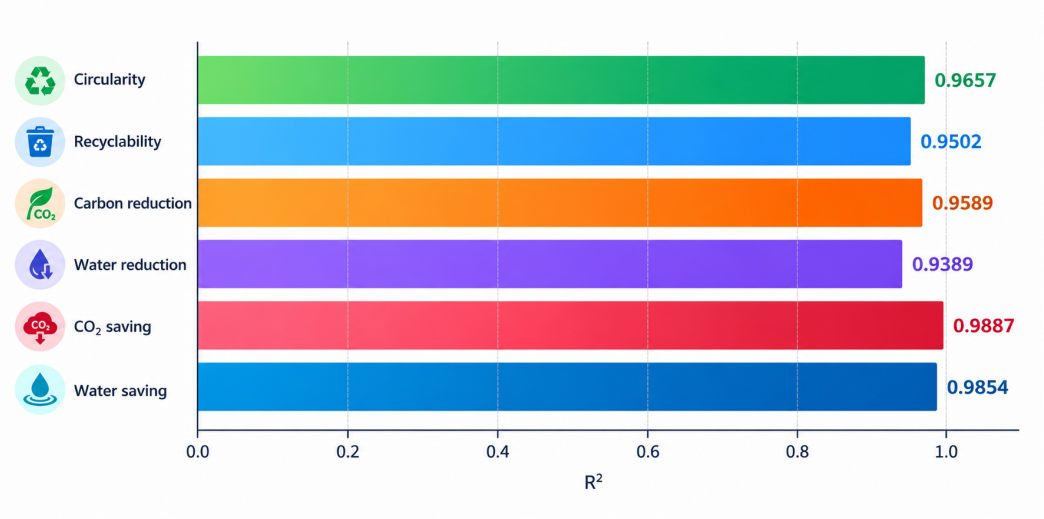


Figure 4:Regression Evaluation Results – Development Dataset

These results support H1 within the specified development data set only. They suggest that the implemented pipeline can be fitted to the available synthetic target relationships but not that the model will generalize to real textile waste or that the predicted environmental values represent measured physical savings.

Target	MAE	RMSE	R^2
Circularity score	2.429	3.043	0.9657
Recyclability score	2.459	3.088	0.9502
Carbon reduction (%)	2.397	3.009	0.9589
Water reduction (%)	2.464	3.084	0.9389
CO_2 saving (kg)	4.456	6.110	0.9887
Water saving (litres)	310.365	437.567	0.9854

Table 9 .a inability regression evaluation results

The highest R^2 value is achieved for CO_2 saving, followed closely by water saving and circularity. The reduction of water, on the contrary, has the lowest R^2 value among the six targets, although it is relatively high within the current development evaluation. The magnitude of the error should be interpreted on the unit of a particular target, e.g., a water-saving MAE of 310.365 litres cannot be compared directly to a circularity-score MAE of 2.429.

The paper explicitly warns that one should treat the current synthetic data as a technology-workflow proof-of-concept and not as an indication of realistic market or environmental accuracy. This warning has been kept here to avoid overstating the results of this study.

6.2 Product and Technical Results

The implementation resulted in the developed sustainability decision-support architecture that includes assessment, prediction, scenario simulation, history, dashboard analytics, impact-factor management, model management, and integration outputs. The model loader was implemented for native XGBoost JSON import and separate preprocessing pipelines. The API and frontend were connected via the response contract.

The component encompasses more than the mere prediction; it also includes risk flags, explanation texts and factor snapshots for the predicted values. Such features are beneficial for the described use case, as the contextual information is critical for accurate interpretation. For example, if a particular prediction is negative for a specific factor, the corresponding factor snapshot can be used to review the conditions under which the result was obtained.

Capability	**Implemented outcome**
Sustainability prediction	Six environmental/circular regression outputs.
Sustainability grade	Separate classifier using 18-feature representation.
Scenario simulation	Five recovery/disposal pathways compared.
Risk assessment	Risk flags and data-quality status.

Explainability	Main reason, supporting reasons, negative factors and suggestions.
History	Persisted prediction records with filters.
Dashboard	Totals, distributions, trends and analytics.
Impact factors	Versioned records with snapshots.
Model management	Dataset upload, registry and activation workflow.
Integration	Downstream-ready sustainability payload.

Table 10 Implemented product and technical capabilities

6.3 User or stakeholder validation

Formal stakeholder validation with a sampled industry population is, outside the current evidence set. The available validation is therefore interpreted as workflow validation: the component was integrated into the seller-side workflow the sustainability grading call was connected to the uploaded textile analysis flow and the resulting sustainability values were passed to the marketplace prediction function.

A future stakeholder study should evaluate whether sellers, recyclers and buyers understand the grade, explanation and scenario ranking, whether the outputs change routing decisions and whether the interface communicates uncertainty appropriately. Such a study should be separated from model accuracy evaluation.

6.4 Integration results

The seller workflow shows how the integration is meant to work. After the application analyzes a textile image it determines the fabric type and any health-related details. Then it uses the item's attributes to request a sustainability grading. Once a grade is returned the six environmental outputs are sent to the marketplace prediction system. The listing workflow can then save the item along with its sustainability-related metadata.

This integration proves that Sustainability Intelligence isn’t something kept in a research notebook. Its results are available right when a marketplace decision needs to be made. The architecture supports the idea of connecting intelligence, with real commercial choices. It brings research into use.

6.5 Summary of findings

- The six sustainability predictors got R squared values between 0.9389 and 0.9887 on the development evaluation dataset that's available.
- The system has preprocessing steps for regression and for classification.
- Using XGBoost JSON files helps reduce reliance on model saving methods.
- Services, for scenario, risk and explanation add value to the machine learning results and help in decision making.
- The sustainability results can be used by parts of the marketplace.
- The data that is being used is made up. The results show that the pipeline works but are not real environmental results.

7 DISCUSSIONS

7.1 Interpretation of findings

The regression metrics that have been reported show that the chosen modelling approach can recreate the target relationships that exist in the development data. I think this is valuable as an engineering milestone because the pipeline covers the journey, from structured input through preprocessing, prediction, grading and the API response.

However, the interpretation must stay within the limits of the dataset. Synthetic data that is created from rules can give high R^2 values because the model learns patterns that are already built into the generation logic. A model that does well on data may behave very differently when it faces mixed fibres, contamination, regional differences, measurement noise and real recovery outcomes. The reliable conclusion we can draw is that the component has reached a technically coherent and testable machine-learning-integrated prototype stage.

7.2 Comparison with literature baselines and existing solutions

The component expands what current textile AI methods can do by connecting factors to what happens in the marketplace. Research about sorting textiles shows how helpful it is to recognize materials [11, 12, 18, 16, 13] . Work on sustainability and life cycle assessments shows how important it is to have environmental details and good quality data [4, 8, 17, 10]. Studies on the economy focus on keeping materials in use and making more circular value [2, 5].

The main point is that this is about structure and bringing things together much as it is about algorithms. The new idea is not that XGBoost is a sustainability tool; the new idea is using a structured machine learning part inside a larger textile marketplace process along, with keeping track of different versions of factors comparing different situations checking risks and connecting with what happens later.

7.3 Research contribution

The research contribution of Member 02 is a sustainability-intelligence methodology that brings circularity and environmental indicators to life within a marketplace

workflow. The contribution includes a feature and target structure, a multi-stage prediction pipeline, a clear line between environmental estimates and formal life cycle assessment and an audit-friendly factor-snapshot mechanism.

A second contribution is honesty about evidence. Because the current dataset is synthetic the thesis makes a difference, between development validation and real-world environmental validation. This difference strengthens the research credibility of the component because it stops model-fit numbers from being shown as proof of actual environmental impact.

7.4 Product contribution

I see that from a product perspective Sustainability Intelligence turns analysis, into marketplace information that anyone can act on. A seller can receive not a sustainability grade but also environmental indicators, scenario comparisons, risk conditions and an explanation. Downstream modules can use Sustainability Intelligence outputs instead of scraping a user interface.

The product architecture also supports model evolution. A new trained model can be. Activated without changing the frontend contract. Impact factors can be. Historical predictions keep the factor snapshot that was used at the time of inference.

7.5 Limitations and threats to validity

The main limitation is the use of training and evaluation data. This affects validity because real textile waste can be very different from the generated data. A second limitation is that environmental factors are based on estimates and assumptions than actual measurements for each item. A third limitation is that there has been no validation with stakeholders. A fourth limitation is that the current prototype does not include production-grade authentication and authorization.

There are also modeling threats. The classifier depends on the regressors so errors, in regression can carry over into classification. Using data to generate targets may create

correlations that do not exist in real-world data. Location and industry features might end up acting as proxies for variables if data collection isn't carefully designed. These risks need to be assessed before the system moves into use.

7.6 Implications

For research the component shows a way to move from circular economy ideas to a software tool that uses machine learning. For engineering it proves the benefit of keeping model predictions separate, from risk, explanation, scenario and integration services. For the Sri Lankan textile sector the biggest future effect is the chance to build a dataset that can help make better environmental and business choices.

The study also suggests that sustainability scores need to have background. A score should not be seen as an environmental fact; it should be connected to the model version the factors used the quality of the data and the assumptions that were made. This is very important when environmental results are used in business rankings or pricing systems.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis showed how the Member 02 Sustainability Intelligence part was designed, built and tested. This part takes eight textile details and changes them into six predictions about the environment and circular economy. It also creates four indicators and a grade for sustainability. It adds features like comparing different situations checking risks giving reasons showing details keeping records and connecting with other parts of the system. The results from the tests show that the model works well with the synthetic data set. The R squared values are between 0.9389 and 0.9887 for the six predictions. The main point is not that these numbers show environmental accuracy. Instead the work shows an repeatable technical process and explains what evidence is needed for the next step, in testing.

8.2 Achievement of objectives

Objective	Achievement	Evidence
Identify sustainability inputs	Achieved	Eight raw features defined and implemented.
Develop environmental prediction	Achieved	Six XGBoost regression estimators.
Develop sustainability grade	Achieved	Separate classifier with 18-feature representation.
Integrate with API	Achieved	FastAPI prediction/integration workflow.
Provide explainable decision support	Achieved	Risk, explanation and scenario services.
Evaluate models	Achieved	MAE, RMSE and R^2 reported for six targets.
Define real-data validation path	Achieved	Limitations and future data strategy documented.

Table 11 .Target users and value proposition

8.3 Individual research and product contributions

The individual contribution mixes sustainability research, feature and target definition, dataset preparation, model training and evaluation model artifact handling, API

integration, frontend integration, testing and documentation. The contribution is part of the team platform but the contribution can still be seen as a separate piece because the contribution's input/output contract and the contribution's model pipeline are clear.

The product contribution is a sustainability decision layer that can run as a service and can also be used inside the seller listing workflow. The research contribution is a methodology that turns textile characteristics into circularity and environmental decision-support outputs while the research contribution keeps prototype evidence separate, from validated environmental measurement.

8.4 Future directions

1. Organize real observations of Sri Lankan textile waste, including different fabric types, conditions and sources from industries.
2. Adjust synthetic target-generation rules by using reliable environmental inventory data that has been verified.
3. Perform testing, training and validation with strict leakage controls and detailed subgroup analysis.
4. Include confidence calibration, uncertainty intervals and thorough residual and error analysis.
5. Check scenario recommendations against outcomes in reuse, repair, upcycling and recycling activities.
6. Implement authentication, authorization and secure controls for system access.
7. Connect outputs with pricing tools and buyer recommendation systems.
8. Enable automated model cards, experiment tracking and strong model governance practices.
9. Create procedures for managing environmental factors and approve model versions through structured governance.
10. Assess usability, with textile manufacturers, recyclers, designers and marketplace operators.

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GLOSSARY

Term	Definition
Circularity score	A modelled indicator representing the estimated circular value of a textile item within the defined development methodology.
Environmental factor	A parameter used to translate material or recovery information into estimated environmental impact or saving.
Impact-factor snapshot	The factor values stored with a prediction so that historical results remain auditable.
Model artifact	A saved trained model or preprocessing object required for inference.
Sustainability grade	A classification output summarizing the sustainability characteristics represented by the model inputs and predicted indicators.
Scenario simulation	Comparison of alternative recovery/disposal pathways such as reuse, repair, upcycle, recycle and discard.
Synthetic dataset	Data generated according to predefined rules for development or pipeline testing rather than collected from real-world observations.
Training/inference consistency	The requirement that production inputs are transformed in the same way as training inputs.
XGBoost	An implementation of scalable gradient-boosted decision trees used here for structured regression and classification.

APPENDICES

Appendix A Research instruments or extended results

Stage	Specification
Raw inputs	fabric_type; weight_kg; health_score; repairability; condition; district; province; industry_type
Regression outputs	circularity_score; recyclability_score; carbon_reduction_percent; water_reduction_percent; co2_saved_kg; water_saved_liters
Derived outputs	energy_saved_kwh; landfill_diverted_kg; sdg_impact_score; economic_impact_score
Classifier inputs	8 raw features + 6 regression outputs + 4 derived indicators
Regressors	Six XGBRegressor models loaded from native JSON artifacts
Classifier	One XGBClassifier loaded from native JSON artifact
Preprocessing	Separate fitted regression and classification ColumnTransformers
Encoding	Grade label encoder persisted separately

Table 12. Extended model specification

Appendix B Academic integrity and AI use disclosure

Week	Date	Individual contribution
1	01/04–07/04	Discussed sustainability requirements; identified environmental indicators; discussed CO_2 and water-saving predictions; contributed to module planning.
2	08/04–14/04	Researched textile environmental impacts and sustainability; identified sustainability metrics; shared findings.
3	15/04–21/04	Reviewed sustainability papers; researched CO_2-saving and water-saving

		approaches; collected references.
4	22/04–28/04	Searched sustainability datasets; compared data sources; identified prediction features.
5	29/04–05/05	Organized sustainability dataset; checked quality; identified missing/invalid values; prepared initial data.
6	06/05–12/05	Cleaned sustainability data; prepared numerical features and target variables; prepared training/testing data.
7	13/05–19/05	Created Colab environment; prepared ML libraries; built training pipeline; started baseline training.
8	20/05–26/05	Trained initial sustainability model; recorded performance; evaluated predictions; identified weaknesses.
9	27/05–02/06	Tested ML algorithms and features; compared performance; documented results.
10	03/06–09/06	Tuned model parameters; improved preprocessing; retrained and compared results.
11	10/06–16/06	Analysed model errors; investigated important features; checked prediction quality; identified improvements.
12	17/06–23/06	Selected sustainability models; saved trained models; prepared model files; documented performance.
13	24/06–30/06	Continued model and API preparation; tested different inputs; prepared

		API requirements; documented implementation.
14	01/07–07/07	Integrated sustainability model with API; tested model loading and prediction requests; fixed integration issues.
15	08/07–14/07	Integrated Sustainability Analysis into seller workflow; tested frontend/backend communication and result display.
16	15/07–21/07	Tested sustainability analysis workflow, CO_2 and water-saving predictions and sustainability scoring.
17	22/07–28/07	Reviewed model accuracy; analysed incorrect predictions; identified improvements; updated documentation.
18	29/07–04/08	Gathered Sri Lankan apparel/textile information; reviewed environmental reports and sustainability practices.
19	05/08–11/08	Continued research writing; documented dataset and methodology; added experimental results.
20	12/08–18/08	Reviewed model accuracy; prepared evidence; tested complete sustainability workflow; prepared presentation content.
21	19/08–22/08	Finalized current evaluation; prepared viva explanation; organized evidence; continued final testing and bug fixing.

Table 13 .Member 02 contribution timeline

Appendix C: Verification Checklist

1. Start the FastAPI backend and confirm the health endpoint.
2. Start the React frontend and verify route navigation.
3. Submit a valid sustainability assessment.
4. Confirm six sustainability outputs are returned.
5. Confirm the sustainability grade is returned.
6. Confirm risk and data-quality information is visible.
7. Run scenario simulation and confirm all five scenarios are displayed.
8. Confirm the best scenario is highlighted.
9. Confirm the prediction is persisted in history.
10. Confirm dashboard analytics reflect saved predictions.
11. Verify the factor version and snapshot are retained.
12. Verify the integration payload contains the expected sustainability fields.
13. Upload a valid dataset through the ML management page.
14. Verify dataset metadata and validation errors.
15. Load a registered model and verify its version metadata.
16. Run the frontend production build.
17. Run backend and frontend automated tests where available.
18. Record actual screenshots from the tested build for the final submission evidence.

Appendix D: Evidence and Research Integrity Note

The quantitative results in this thesis come from the project documentation and evaluation records related to the Sustainability Intelligence component. The current development dataset is synthetic and created using rules. There is no claim that the reported R^2 MAE or RMSE values reflect certified performance in the Sri Lankan textile sector. Real system screenshots must be taken from the tested build and should not be fabricated for the thesis.

When a figure or screenshot is required by the assessment it should only be included after confirming that the user interface shown matches the implementation described in this thesis. Any future experiments using data should replace the development-validation interpretation with a clear experimental protocol. This includes a dataset description, a defined data split strategy a comparison, with baseline models and an analysis of uncertainty.

Appendix E: Detailed API and Data Contract

This appendix records the API responsibilities that are written down for the Sustainability Intelligence component. The goal is to make the limits of the implementation clear and to help reproducibility. The endpoint names that are written below describe the EcoThread Intelligence implementation that has been documented. In places where the larger seller workflow calls a ML-service grading function that difference is kept separate instead of being put into one made-up endpoint.

The API agreement keeps validation separate from model inference. Keeps prediction separate, from storage. This means that a future model can be changed without having to redesign the React interface. It also means that modules that run later in the marketplace can use a JSON representation of sustainability outputs.

Endpoint	Method	Primary responsibility	Expected evidence
GET /api/health	GET	Check API process availability independently of MongoDB.	HTTP 200 health response.
POST /api/sustainability/predict	POST	Validate assessment, predict, enrich, persist and return result.	Prediction response + persisted record.
POST /api/sustainability/simulate	POST	Compare reuse, repair, upcycle, recycle and discard.	Scenario rows + best action.
GET /api/sustainability/history	GET	Retrieve persisted predictions with optional filters.	History list.
GET /api/sustainability/dashboard-summary	GET	Aggregate totals, distributions and trends.	Dashboard JSON.
GET /api/reports/sustainability-summary/{prediction_id}	GET	Create report-ready summary.	Summary JSON.
GET /api/integration/sustainability-output/{prediction_id}	GET	Expose downstream sustainability output.	Integration JSON.
GET /api/impact-factors	GET	List active environmental factors.	Factor records.
POST /api/impact-factors	POST	Create a factor record.	Created factor.
PUT /api/impact-factors/{factor_id}	PUT	Update factor values/version metadata.	Updated factor.
DELETE /api/impact-factors/{factor_id}	DELETE	Soft-delete a factor.	Inactive factor state.
POST /api/datasets/upload	POST	Upload and validate CSV data.	Dataset metadata.
GET /api/datasets	GET	List uploaded datasets.	Dataset list.
GET /api/models	GET	List registered model versions.	Model registry list.
GET /api/models/active	GET	Return active trained model metadata.	Active model.
POST /api/models/train	POST	Train models from an uploaded dataset.	Training/registry result.
GET /api/models/{model_version}/metrics	GET	Return evaluation metrics.	Metric object.
POST /api/models/{model_version}/activate	POST	Activate a registered model version.	Activation result.

Appendix F: Model Pipeline Pseudocode and Decision Logic

The following pseudocode expresses the implemented decision sequence without replacing the actual source code. It is included to make the research method understandable to a reader who is not inspecting the repository.

INPUT: fabric_type, weight_kg, health_score, repairability, condition, district, province, industry_type

1. Check the request. Make sure the categories are all in the same format.
2. Create a data frame with eight features that are not changed yet.
3. Use the preprocessor that was already set up to change the data frame.
4. For each of the six things we are trying to predict use the XGBoost regressor that is set up for that.
5. Keep the six predicted values that're about the environment and the circular economy.
6. Figure out the values for energy saved in kilowatt hours waste diverted from landfills in kilograms the score for development goals and the score for economic impact based on the decision support rules that are in place.
7. Put together the eight features the six predicted values and the four calculated indicators into one data frame.
8. Use the classifier preprocessor that is made for this to change the data frame with eighteen features.
9. Use the XGBoost grade classifier. Find out what the grade label is.
10. Make sure the numbers are within limits and are in the right format.
11. Find out the risk flags and the status of how good the data's
12. Make an explanation list the reasons that support it note any negative factors and give some ideas on how to make things better.
13. Look at recovery options by checking how good they are for the environment how easy they are to do how hard they are and how much risk they have.
14. Save the result along with the model and the impact factor information..
15. Send the result back to the front end and make the data available for the next step, in the process.

The model is specifically trained and run using predictions from a regressor of actual real-world values. This matters because real-world values are not available when the model is being used. If the model was trained using values but used with predicted values the data it was trained on and the data it sees later would be different. The way it is set up makes sure it uses the kind of information that is available when it is actually being used.

Decision stage	Information used	Decision-support role
Input validation	Eight raw features	Reject invalid or incomplete requests.
Regression	Eight raw features	Estimate six sustainability outputs.
Derived indicators	Six outputs + item context	Create additional decision signals.
Classification	Eighteen-feature representation	Generate sustainability grade.
Risk	Condition, contamination, distance, quality and other checks	Indicate reliability/practical concerns.
Scenario engine	Environmental benefit + feasibility + difficulty + risk	Rank alternative recovery pathways.
Explanation	Prediction + decision context	Communicate why the result was produced.
Integration	Structured sustainability fields	Support pricing and marketplace modules.

Appendix G: Extended Test Case Catalogue

The test catalogue below is made to act as a checklist, for the final implementation. The expected results show how the system should behave not that every test has been run in this document. Before you submit, capture execution evidence from the running build especially when the assessment asks for screenshots or logs.

ID	Test area	Procedure	Expected outcome
TC-01	API health	Start backend; call /api/health.	API reports healthy process state.

TC-02	Valid assessment	Submit complete eight-feature request.	HTTP success and six sustainability outputs.
TC-03	Missing feature	Remove a required input.	Validation error with safe message.
TC-04	Invalid weight	Send negative/invalid weight.	Validation or safe lower-bound handling.
TC-05	Unknown fabric	Use an unsupported fabric category.	Controlled validation/risk handling.
TC-06	High contamination	Use high contamination condition.	Risk service identifies elevated risk.
TC-07	Long distance	Use high transport distance where supported.	Risk/decision logic reflects transport concern.
TC-08	Scenario comparison	Run all five scenarios.	Rows for reuse, repair, upcycle, recycle and discard.
TC-09	Best scenario	Use a healthy, repairable item.	Repair/reuse pathway can rank highly where logic supports it.
TC-10	Low-quality item	Use poor health/condition.	Scenario engine can downgrade reuse/repair feasibility.
TC-11	Prediction history	Save then query a prediction.	Record appears with metadata.
TC-12	Factor snapshot	Change active factor after a prediction.	Historical prediction retains its previous snapshot.
TC-13	Dashboard	Create several predictions and request summary.	Totals/distributions/trends are returned.
TC-14	Report summary	Request report for a valid prediction ID.	Structured report-ready JSON.
TC-15	Integration payload	Request downstream output.	Expected sustainability fields are returned.
TC-16	Dataset upload valid	Upload valid CSV with required columns.	Dataset accepted and metadata stored.
TC-17	Dataset upload invalid	Upload wrong file/missing columns.	Controlled validation failure.

TC-18	Model registry	List registered models.	Model versions and metadata displayed.
TC-19	Model activation	Activate a registered model.	Selected version becomes active.
TC-20	Missing trained artifact	Run auto mode without trained artifacts.	Placeholder fallback is used if configured.
TC-21	Frontend API mapping	Submit assessment through UI.	Returned values display in correct cards/sections.
TC-22	Responsive scenario UI	Open scenario page at laptop width.	No unintended horizontal overflow.
TC-23	Build verification	Run production build.	Frontend compiles successfully.
TC-24	Regression evaluation	Run evaluation script.	MAE, RMSE and R^2 are produced.
TC-25	Classification evaluation	Run classification evaluation.	Accuracy/precision/recall/F1/confusion matrix are produced.

Appendix H: Evaluation Interpretation Guide

The model results should be carefully considered in terms of scale how the target is defined and where the data came from. A high R^2 value means the model captures a part of the variation in the target variable within the specific dataset and split used.. It does not mean the predicted value is something that was actually measured in the real world. The MAE and RMSE values need to be understood in the units of the target variable. These metrics should be compared to baseline models before assuming the model is accurate in practice.

For the development dataset all six R^2 values are above 0.93. This is a signal from an engineering perspective. It shows the pipeline can effectively learn the patterns and relationships in the data. However the dataset is synthetic which raises concerns about external validity. Real-world textile waste will have measurement errors, mixed materials, unknown contaminants, regional differences and operational limits that are not present in the simulated data.

Because of this the next evaluation step should include least six key components: (1) a separate real dataset that was not used in training (2) a baseline model for comparison (3) analysis broken down into subgroups, (4) examination of how errors are distributed, (5) an assessment of uncertainty or confidence intervals and (6) direct comparison with actual recovery outcomes from real operations.

If real data is limited techniques like crossvalidation and sensitivity analysis can help strengthen the evidence.. These methods cannot replace the need, for external validation. Real-world performance must be tested on data that truly reflects the conditions the model will face in practice.

Current evidence	**What it supports**	**What it does not support**
R^2/MAE/RMSE on synthetic data	Pipeline can fit development targets.	Certified environmental accuracy.
API tests	Software contract behaves as designed.	Real-world stakeholder acceptance.
Frontend tests/build	UI and integration are technically coherent.	Environmental impact claims.
Factor snapshots	Historical assumptions are auditable.	Correctness of the factors themselves.
Scenario ranking	Decision logic is executable and explainable.	That the selected pathway always has the lowest real impact.

Appendix I: Suggested Final Evidence Capture Plan

The final thesis needs to have screenshots taken from the working version of the system. The following evidence plan connects screenshots to the statements made in the sections. This appendix does not make up any images on purpose.

Evidence 1. Sustainability Assessment form: take a picture of the form that is fully filled out. Show the fabric type, weight, condition, repairability, location and industry fields.

Evidence 2. Sustainability result: take a picture of the results that include the grade, circularity and recyclability values, carbon and water reductions, CO_2 and water savings, confidence and data-quality indicators and the recommendation.

Evidence 3. Scenario Simulation: take a picture of the five scenarios and the recommended action that is highlighted.

Evidence 4. Prediction History: take a picture of the saved predictions and the filters that are applied.

Evidence 5. Dashboard: take a picture of the totals, trends and distributions after test predictions.

Evidence 6. Impact Factors: take a picture of the view for managing environmental factors.

Evidence 7. ML Models: take a picture of the dataset and model registry and the active model information if it is available.

Evidence 8. Seller integration: take a picture of the seller process after the textile AI analysis and before the product is published. Show the sustainability results that are shared in the marketplace process.

Each screenshot needs a number in the caption. Must be mentioned in the text, around it. The screenshots must be clear cropped to the parts of the application and free of personal information that is not needed.

Evidence	**Supports**	**Suggested chapter**
E1	Assessment inputs and UI requirements	4, 5
E2	Prediction and grading output	5, 6
E3	Scenario decision support	4, 5, 6
E4	Persistence/history	5, 6
E5	Analytics	5, 6
E6	Factor versioning	4, 5
E7	Model lifecycle	5
E8	Marketplace integration	5, 6, 8